Status	Finished
Started	Wednesday, 21 October 2026, 11:33 PM
Completed	Wednesday, 21 October 2026, 11:35 PM
Duration	2 mins 17 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Linear Search
b. Binary Search
c. Bubble Sort
d. Merge Sort

The correct answer is: Merge Sort

Question **2**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It compares adjacent elements and swaps them if necessary
b. It builds a sorted array one element at a time
c. It divides the list into two halves, sorts each half, and then merges them
d. It repeatedly finds the minimum element and moves it to the sorted part of the list

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **3**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Quick Sort
b. Merge Sort
c. Insertion Sort
d. Bubble Sort

The correct answer is: Bubble Sort

Question **4**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Merge Sort
b. Bubble Sort
c. Linear Search
d. Binary Search

The correct answer is: Bubble Sort

Question **5**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Merging
b. Searching
c. Complexity
d. Sorting

The correct answer is:
Complexity

Question **6**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It divides the list into sublists
b. It recursively sorts subproblems
c. It uses the heap data structure
d. It bubbles up the largest element in each pass

The correct answer is: It bubbles up the largest element in each pass

Question **7**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Extraction
b. Distribution
c. Selection
d. Insertion

The correct answer is:
Insertion

Question **8**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

a. Searching for data in a list
b. Arranging data in a particular format
c. Inserting data into a list
d. Deleting data from a list

The correct answer is: Arranging data in a particular format

Question **9**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Merge sort
b. Heap sort
c. Quick sort
d. Selection sort

The correct answer is:
Merge sort

Question **10**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

a. arrange()
b. order()
c. sort()
d. sorted()

The correct answer is: sorted()

Question **11**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 40,70,80,90,120,150,200,300,400,470
b. 80,150,200,470,40,90,300,400,70,120
c. 200,470,80,150,40,90,300,400,70,120
d. 40,80,90,150,200,300,400,470,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **12**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

a. Sorting
b. Searching
c. Merging
d. Rearranging

The correct answer is: Sorting

Question **13**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Dividing the list into halves
b. Bubbling up the largest element to its correct position
c. Sorting the entire list in one pass
d. Bubbling up the smallest element

The correct answer is: Bubbling up the largest element to its correct position

Question **14**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. Duplicates
b. Selection
c. All of the above
d. Searching

The correct answer is: All of the above

Question **15**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Bubble Sort
b. Merge Sort
c. Linear Search
d. Binary Search

The correct answer is: Merge Sort